S475/1
SUBSIDIARY
MATHEMATICS
PAPER 1
JULY/AUG 2026
2¼ HOURS

PROVINCIAL EXAMINATIONS AGENCY (PEA)
COUHEIA MOCK EXAMINATIONS 2026
Uganda Advanced Certificate Of Education
SUBSIDIARY MATHEMATICS
Paper 1
TIME: 2¼ HOURS

INSTRUCTIONS TO CANDIDATES:

This paper consists of **Two** *sections;* **A** and **B.** *It has* **six** *examinations items.*

Section **A** *has* **two compulsory** items.

Section **B** *has* **two parts**. *Answer one item only from each part*

Answer a total of **four** *items.*

Any additional item(s) answered will **not** *be scored.*

All *answers* **must** *be written in the answer booklets/sheets provided.*

All *working* **must** *be shown clearly.*

Begin each answer on a fresh sheet of paper.

Graph paper is provided.

Silent non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

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SECTION A

(Answer all items in this section.)

ITEM 1

The Government of Uganda is extending electricity to several villages to improve access to social services and promote economic development. An electricity substation is located at point **A**, while two villages that require electricity are located at points **B** and **C**. Surveyors establish the positions of the villages relative to a fixed origin O as follows:

$$\overrightarrow{OA} = 2\mathbf{i} + 3\mathbf{j}, \quad \overrightarrow{OB} = 10\mathbf{i} + 9\mathbf{j}, \quad \overrightarrow{OC} = 14\mathbf{i} + 5\mathbf{j},$$

where one unit represents **1 kilometre.**

The engineer planned to connect each village directly to the substation using transmission lines. To estimate the quantity of materials required, they need to determine the lengths of the transmission lines. It is required that the angle between two transmission lines from the station is less than 45^0 so that they can be installed safely without unnecessary crossing and reduce unnecessary cable bending.

During the design of the transmission line from the substation to village **B**, the engineer modelled its inclination and obtains the equation $4\sin^2\theta - 3 = 0,$

where θ is the angle of inclination of the transmission line measured from the horizontal.

There is need to identify the value θ in the range $0° \leq \theta \leq 360°$. that is suitable for the transmission line from the substation to village **B.**

In the absence of the engineer, you have been requested to offer assistance.

Task

a. Provide the necessary guidance on this project.

ITEM 2

A coffee processing factory purchases coffee beans from farmers, processes them, and prepares them for export. To improve productivity and profitability, the factory management monitors the quantity of processed coffee produced each day.

The quantity of processed coffee, Q tonnes, produced $t$days after the beginning of a production cycle is modelled by

$$Q = x^3 - 9x^2 + 24x + 10, \quad 0 \leq x \leq 7$$

where x is measured in days.

The management is interested in determining the rate at which production changes, the periods when production is highest or lowest.

At the beginning of the processing day ($t = 0$), there are **1,000 bags** of processed coffee in the warehouse. The processing machine operates continuously and produces coffee at a constant rate of **500 bags per hour.**

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As the day progresses, more transport trucks arrive to collect coffee for delivery. Consequently, the rate at which coffee leaves the warehouse increases with time and is given by $D(t) = 300 + 50t$ bags per hour,

where t is the time in hours after processing begins.

One customer requested to send track after 4 hours to pick 1350 bags of coffee. The warehouse manager wants to monitor the stock level to ensure there is enough coffee to meet customer demand.

Hint: At the warehouse, Net Change Rate of stock = Production Rate - Demand Rate

Tasks

Help the manager to

a.
   i. Determine the rate of change of coffee production with respect to time.
   ii. Determine the periods when production is highest or lowest.

b. monitor the stock level by the time the track arrives.

SECTION B

PART I (*Answer **one item only.***)

ITEM 3

The Traffic Police Department in a certain district has observed an increase in road accidents, especially during weekends. To improve road safety, traffic officers conducted a speed-monitoring exercise at a major highway checkpoint. During selected observation periods, they recorded the speeds of vehicles passing through the checkpoint and carried out alcohol tests on drivers.

After collecting the data, the Operations Officer issued the following directives:

i. Drivers whose vehicles do not exceed the average speed recorded during the survey may be pardoned and allowed to proceed without penalty.

ii. If statistical analysis shows that 90% of the vehicles are travelling at speeds greater than 67 km/h, additional traffic officers must be deployed on that highway section to strengthen enforcement and reduce accidents.

The recorded speeds (in km/h) of the 40 vehicles observed are shown below:

24	55	63	34	36	56	40	28	43	66
55	58	72	48	49	50	51	61	52	53
53	54	45	41	56	39	57	57	46	58
59	60	51	62	31	64	65	44	68	47

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In addition, the permit inspection revealed that 60% of the drivers were male. Of the male drivers, 85% possessed valid permits. Of the female drivers, 90% possessed valid permits. All drivers with invalid permits must pay a fine.

The officers also discovered that the probability that a driver tested at the checkpoint is found driving under the influence of alcohol is 0.6. On a certain day, sample of 8 drivers undergone alcohol test.

More alcohol test machines will be provided if the expected number of drunk drivers is more than 4 and that the probability that more than 6 drivers are found drunk exceeds 0.15.

Task

a. Help the Operations officer to know
   i. below what speed should vehicles be pardoned.
   ii. if there is need for more traffic officers.
b. Find the number of drivers who were fined for invalid permits.
c. Advise the Traffic Police on whether more resources should be allocated to alcohol-testing operations at the checkpoint.

ITEM 4

Following a health directive from the Ministry, the Head teacher of a boarding secondary school plans to purchase sanitation supplies for the Girls' and Boys' sections from wholesale shops to minimize costs and improve hygiene standards.

The Girls' Section requires 40 jerrycans of liquid soap, 25 long-handled brushes, and 15 packets of disinfectant. The Boys' Section requires 35 jerrycans of liquid soap, 30 long-handled brushes, and 10 packets of disinfectant.

The costs of the items are shown below:

Item	Cost per Unit (Shs)
Liquid soap (jerrycan)	20,000
Long-handled brush	8,000
Disinfectant (packet)	12,000

To monitor the effectiveness of sanitation measures, the school health committee recorded the number of sanitation-related illness cases reported in the boarding section over six consecutive months.

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Month	Jan	Feb	Mar	Apr	May	Jun
Number of Cases	36	32	25	28	19	21

The committee also conducted a study on student absenteeism due to sanitation-related illnesses. From past records, the probability distribution for the number of students absent on any given day is shown below.

Number of Students Absent (x)	0	1	2	3	4
P(X)	0.20	0.30	0.25	0.15	0.10

The Head teacher needs a report that;

i) indicate the sanitation budget,

ii) analyses a 3-month trends in student health by and estimates cases in July,

iii) estimate the expected daily absenteeism caused by sanitation-related illnesses.

Task

As part of the healthy committee member, prepare the necessary requirements for the head teacher

PART II (*Answer **one item only.***)

ITEM 5

A district hospital is organizing a Community Health Outreach Programme to increase awareness about malaria prevention, immunization, and maternal health services.

On Monday, a village health worker informed 3 households about the programme. Each of the households was requested to inform 3 other households the following day, and this process continued throughout the week. By Sunday, health officials expected that information about the programme would have reached a large number of families in the community.

Among the invited guests are 5 senior doctors and 7 district health officers from different districts.

During the programme:

- The 5 senior doctors are to be seated in a straight row during a panel discussion, with the Chief Medical Officer occupying the seat at the extreme left.
- The 7 district health officers are to be seated around a circular table to discuss strategies for reducing the spread of communicable diseases.

Later, a health advisory committee of 6 members is to be selected from the doctors and health officers to oversee the implementation of health recommendations. To ensure adequate medical expertise, the committee must include at least 2 doctors.

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Meanwhile, the hospital has received donations of medical supplies that need to be packaged for distribution to remote health centres. A health worker has a rectangular piece of cardboard measuring 40 cm by 30 cm and intends to cut equal squares from each corner before folding the sides to form an open box. The hospital requires the box to have a base area of 936 cm^2, but the health worker is unsure of what size to cut.

Task

As a mathematics student assisting the hospital management:

a. Estimate the total number of households that are likely to have received information about the health outreach programme by Sunday.
b. Determine:
   i. The number of different seating arrangements for the senior doctors and district health officers.
   ii. The number of ways the health advisory committee can be selected.
c. Advise the health worker on the size of the squares that should be cut from the cardboard to produce the required box volume for packaging the medical supplies.

ITEM 6

A building company plans to construct a rectangular workers' storage yard. To monitor project expenses, the finance department developed a profit model given by

$$P(x) = x^2 - 16x + 60$$

where x represents the number of work units completed. The management wishes to determine the units at which the company makes neither profit nor loss.

The company also keeps records of construction materials. While calculating material ratios for concrete mixtures, an engineer encountered the expression

$$\frac{1}{2-\sqrt{3}}$$

and needed to simplify it to know how many wheels of sand (a) bags of cement (b) and wheels of stones (c) to use in form $a + b\sqrt{c}$.

To meet increasing demand for housing, the company plans to increase the number of houses completed each month. In the first month, 20 houses were completed. Thereafter, the company completed 5 more houses each month than in the previous month. The manager wishes to know how long it will take them to make 90 houses

Tasks

As a Mathematics student assisting the building company to:

a. determine the break even point.
b. obtain the concrete mixture ratios.
c. find the length of time needed to curb the demand for housing.

END

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